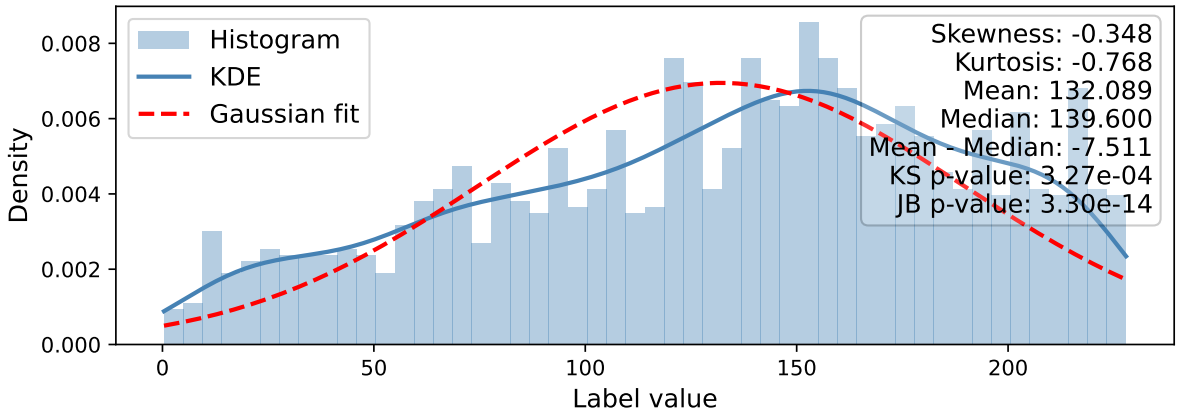
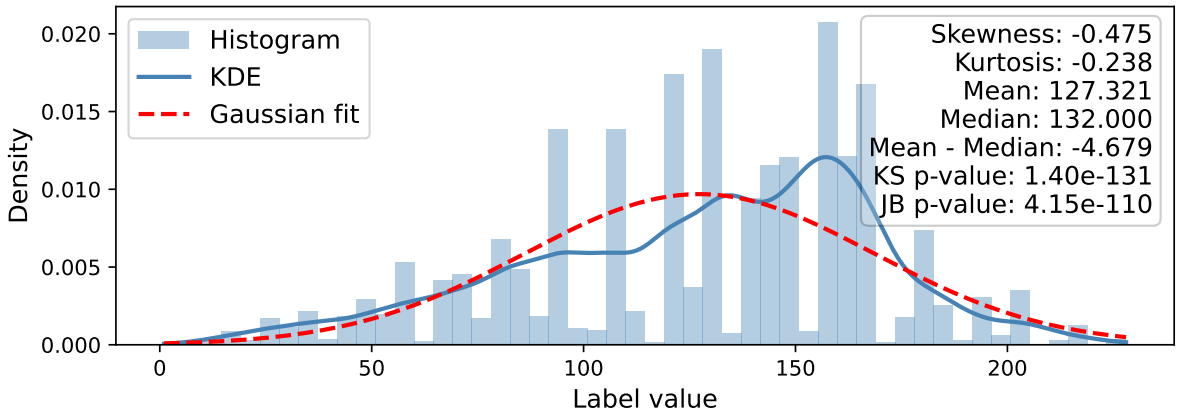
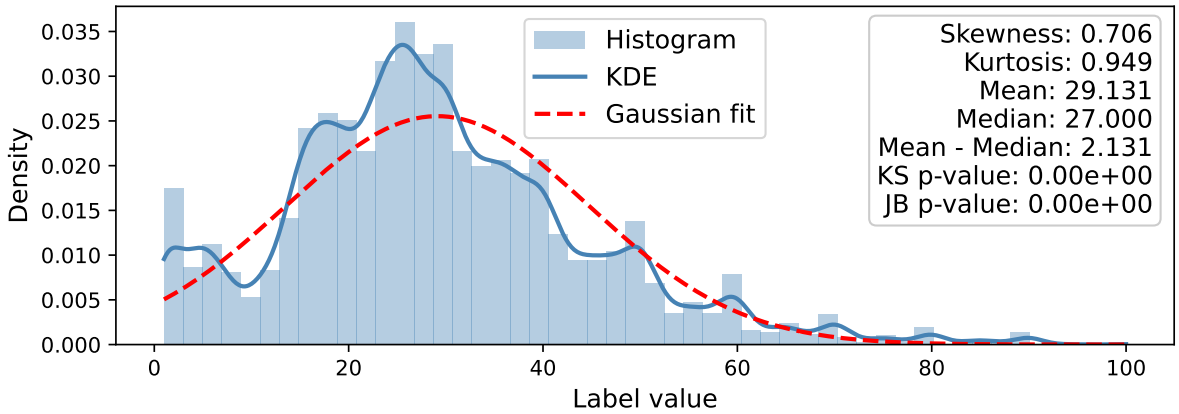


(a) DHA

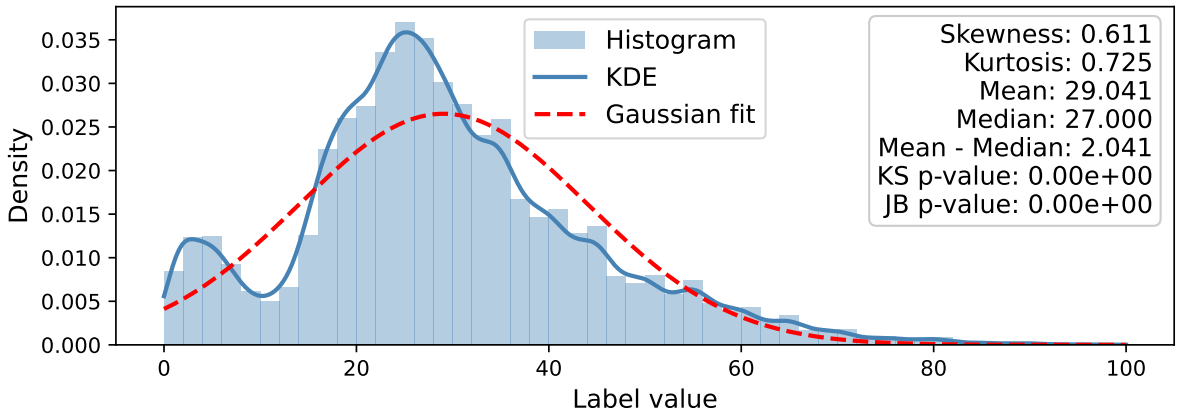


(b) RSNA

Figure 1: Label distributions of the **Bone-Age** datasets: **(1a)** DHA and **(1b)** RSNA. Each panel shows the **histogram** (blue bars ■), **KDE-estimated density** (blue curve —), and **Gaussian fit** (red dashed curve - - -). **Summary statistics** include skewness, excess kurtosis, mean, median, their difference (Mean − Median), as well as Kolmogorov–Smirnov (KS) and Jarque–Bera (JB) test p-values for goodness-of-fit against the Gaussian distribution. Both distributions significantly deviate from Gaussian, confirming non-trivial label skewness.

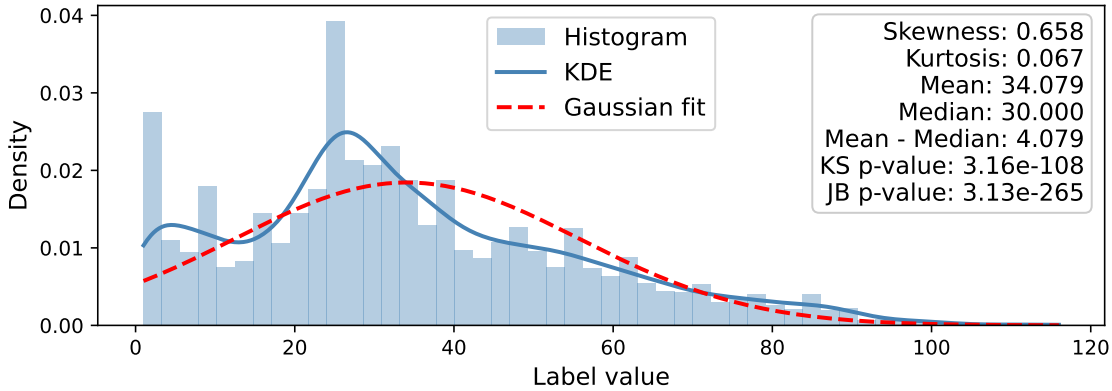


(a) Real Age

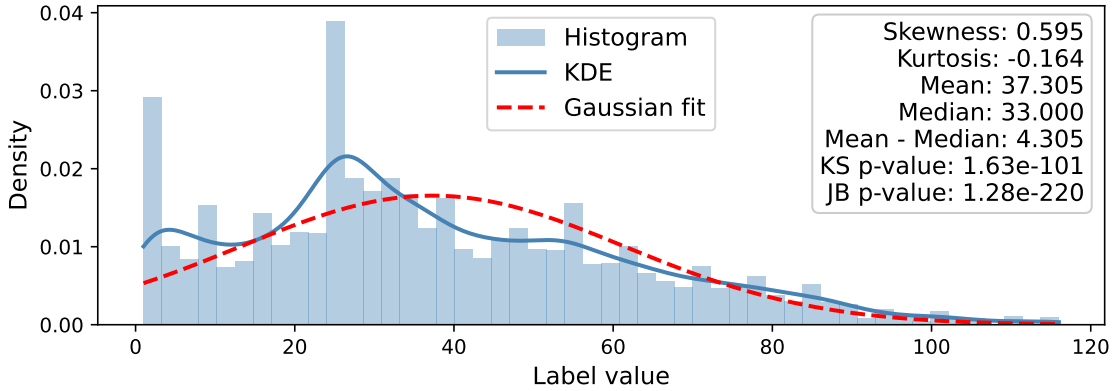


(b) Apparent Age

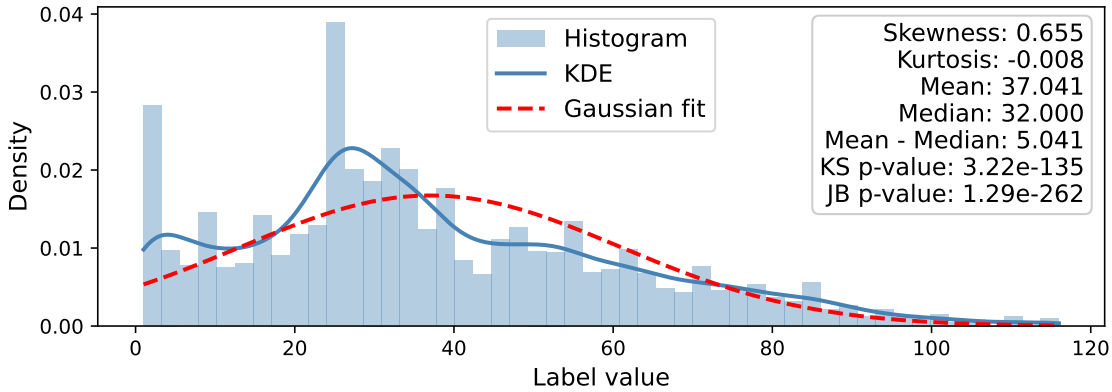
Figure 2: Label distributions of the **APPA-REAL** dataset: **(2a)** Real Age and **(2b)** Apparent Age. Each panel shows the **histogram** (blue bars ▇), **KDE-estimated density** (blue curve —), and **Gaussian fit** (red dashed curve - - -). **Summary statistics** include skewness, excess kurtosis, mean, median, their difference (Mean − Median), as well as Kolmogorov–Smirnov (KS) and Jarque–Bera (JB) test p-values for goodness-of-fit against the Gaussian distribution. Both distributions significantly deviate from Gaussian, confirming non-trivial label skewness.



(a) Gaussian Noise ($\text{std}_{\max} = 30$)

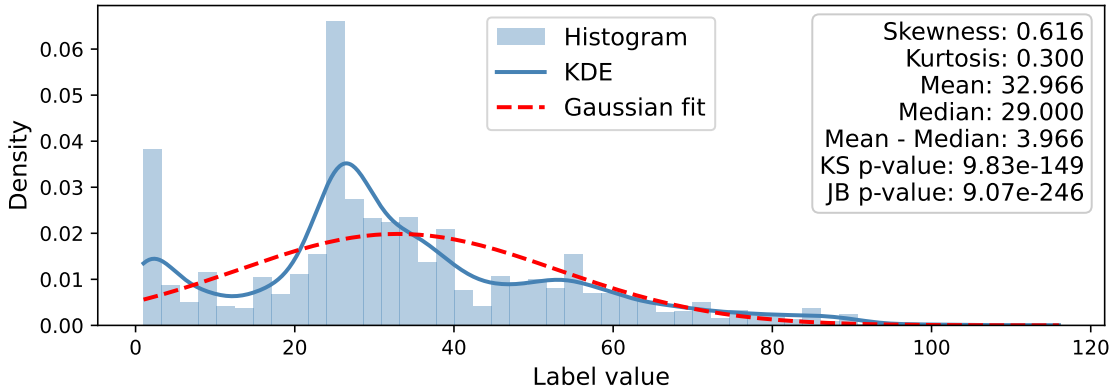


(b) Gaussian Noise ($\text{std}_{\max} = 60$)

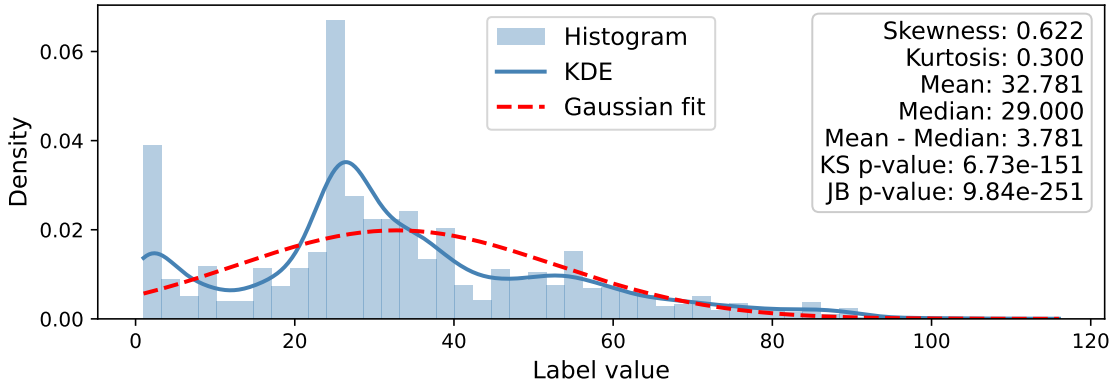


(c) Gaussian Noise ($\text{std}_{\max} = 90$)

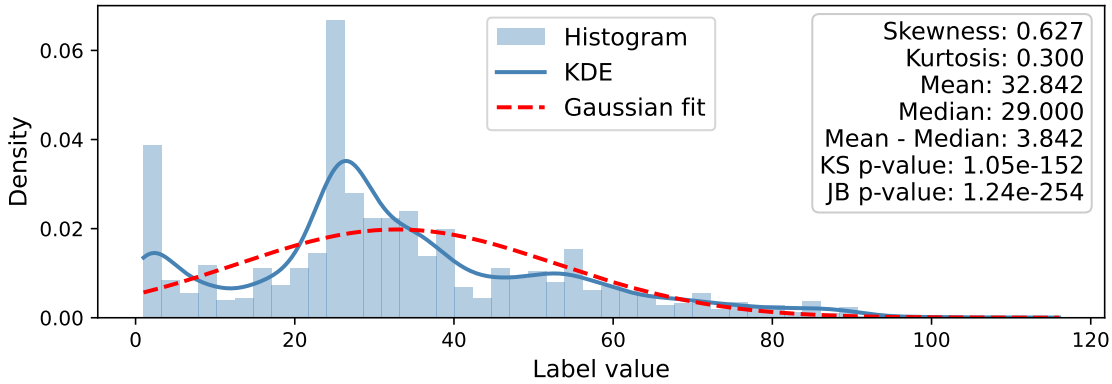
Figure 3: Label distributions of the **UTKFace** dataset under **simulated Gaussian label noise**: **(3a)** $\text{std}_{\max} = 30$, **(3b)** $\text{std}_{\max} = 60$, and **(3c)** $\text{std}_{\max} = 90$. Each panel shows the **histogram** (blue bars ▪), **KDE-estimated density** (blue curve —), and **Gaussian fit** (red dashed curve - - -). **Summary statistics** include skewness, excess kurtosis, mean, median, their difference (Mean − Median), as well as Kolmogorov–Smirnov (KS) and Jarque–Bera (JB) test p-values for goodness-of-fit against the Gaussian distribution. All three noisy label distributions significantly deviate from Gaussian, confirming non-trivial skewness across noise levels.



(a) Random Flip Noise (Error Rate = 0.2)

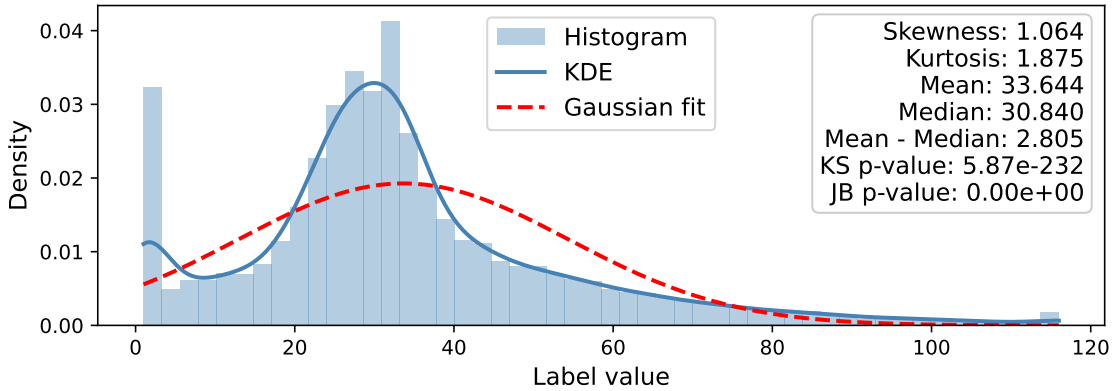


(b) Random Flip Noise (Error Rate = 0.4)

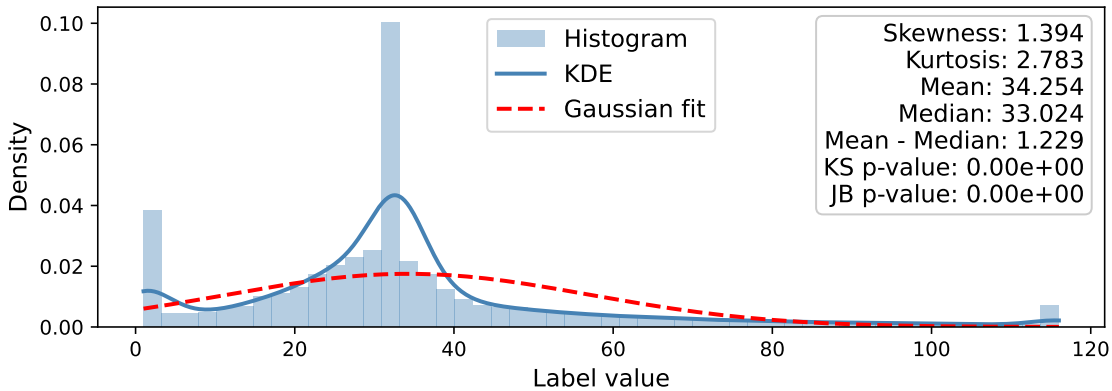


(c) Random Flip Noise (Error Rate = 0.6)

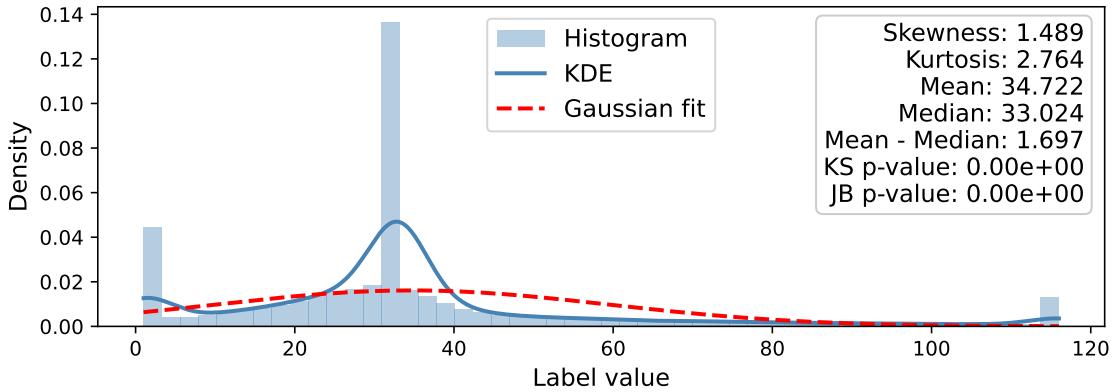
Figure 4: Label distributions of the UTKFace dataset under simulated random flip label noise: **(4a)** Error Rate = 0.2, **(4b)** Error Rate = 0.4, and **(4c)** Error Rate = 0.6. Each panel shows the **histogram** (blue bars ■), **KDE-estimated density** (blue curve —), and **Gaussian fit** (red dashed curve - - -). **Summary statistics** include skewness, excess kurtosis, mean, median, their difference (Mean − Median), as well as Kolmogorov–Smirnov (KS) and Jarque–Bera (JB) test p-values for goodness-of-fit against the Gaussian distribution. All distributions significantly deviate from Gaussian, confirming non-trivial label skewness across noise levels.



(a) Multiplicative Gaussian Noise (std = 0.5)

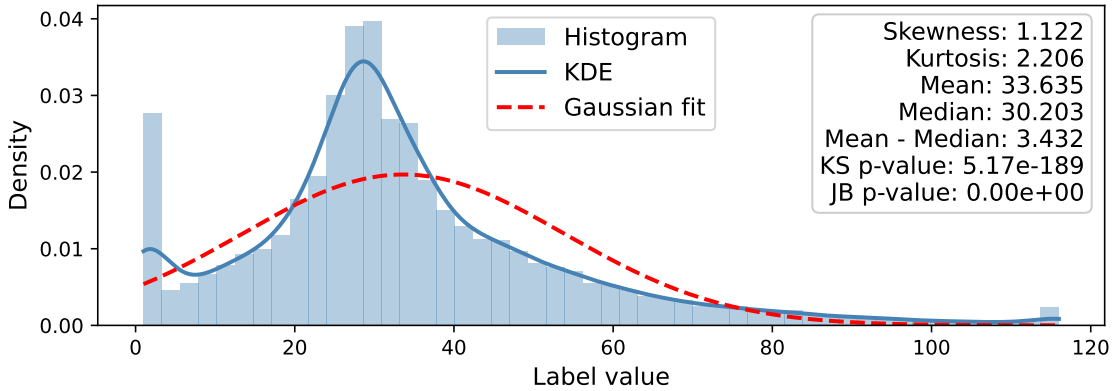


(b) Multiplicative Gaussian Noise (std = 1.0)

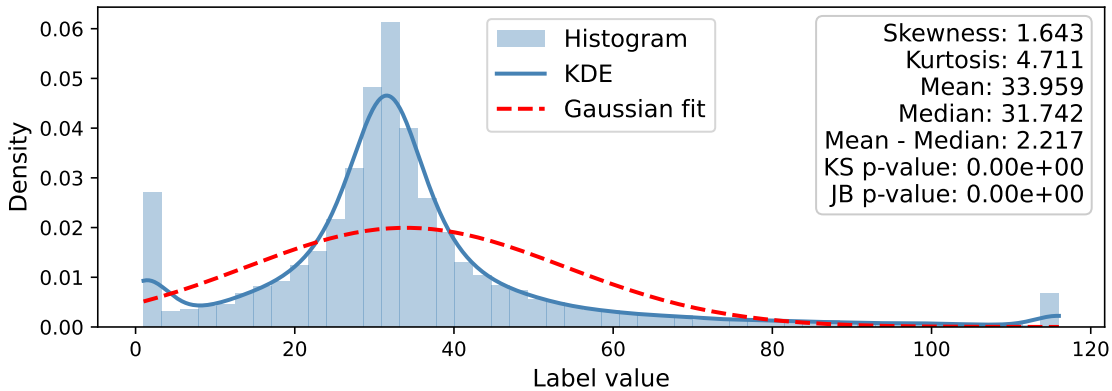


(c) Multiplicative Gaussian Noise (std = 1.5)

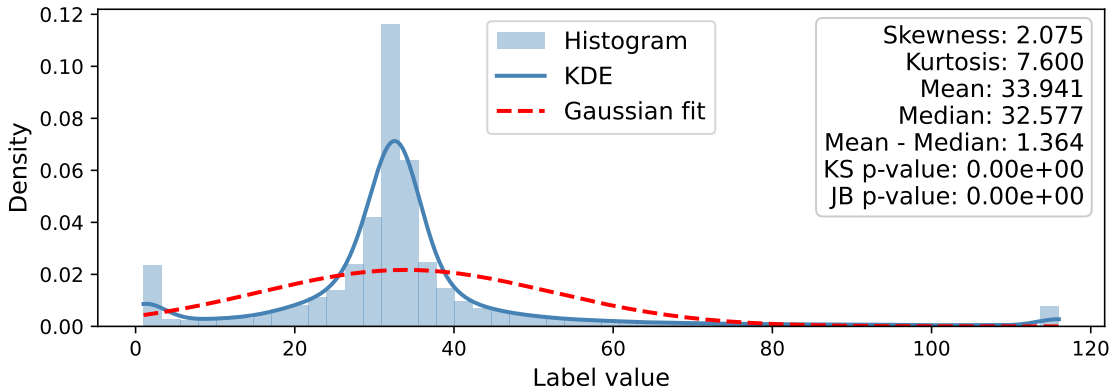
Figure 5: Label distributions of the **UTKFace** dataset under **simulated multiplicative Gaussian noise**: **(5a)** std = 0.5, **(5b)** std = 1.0, and **(5c)** std = 1.5. Each panel shows the **histogram** (blue bars ■), **KDE-estimated density** (blue curve —), and **Gaussian fit** (red dashed curve - - -). **Summary statistics** include skewness, excess kurtosis, mean, median, their difference (Mean − Median), as well as Kolmogorov–Smirnov (KS) and Jarque–Bera (JB) test p-values for goodness-of-fit against the Gaussian distribution. All distributions significantly deviate from Gaussian, confirming non-trivial label skewness across noise levels.



(a) Multiplicative Lognormal Noise (std = 0.5)



(b) Multiplicative Lognormal Noise (std = 1.0)



(c) Multiplicative Lognormal Noise (std = 1.5)

Figure 6: Label distributions of the **UTKFace** dataset under **simulated multiplicative lognormal noise**: **(6a)** std = 0.5, **(6b)** std = 1.0, and **(6c)** std = 1.5. Each panel shows the **histogram** (blue bars ▪), **KDE-estimated density** (blue curve —), and **Gaussian fit** (red dashed curve - - -). **Summary statistics** include skewness, excess kurtosis, mean, median, their difference (Mean − Median), as well as Kolmogorov–Smirnov (KS) and Jarque–Bera (JB) test p-values for goodness-of-fit against the Gaussian distribution. All distributions significantly deviate from Gaussian, with skewness and kurtosis increasing substantially at higher noise levels.